

Techniques of Antidifferentiation: Rational Functions & Partial Fractions

MATH 146 – Calculus II

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Techniques of Antidifferentiation

If $p(x)$ and $q(x)$ are polynomials (so that $\frac{p(x)}{q(x)}$ is a *rational function*), it is always possible (in theory) to evaluate the antiderivative

$$\int \frac{p(x)}{q(x)} dx.$$

However, doing so involves several algebra tricks, and potentially some substitution.

First Idea: Polynomial Division

Example: Evaluate $\int \frac{x^3 + 1}{x^2 - 4} dx$.

Second Idea: Partial Fractions

Example: Evaluate $\int \frac{4x + 1}{x^2 - 4} dx$.

Third Idea: Partial Fractions with Repeated Roots

Example: Evaluate $\int \frac{4x + 1}{x^2 - 4x + 4} dx$.

Fourth Idea: Irreducible Factors

Example: Evaluate $\int \frac{4x + 1}{x^2 + 4} dx$.

Fifth Idea: Several Irreducible Factors

Example: Evaluate $\int \frac{4x + 1}{(x^2 + 4)(x^2 + 1)} dx$.

Sixth Idea: Irreducible Factors and Completing the Square

Example: Evaluate $\int \frac{4x + 1}{x^2 + 2x + 4} dx$.

Seventh Idea: Repeated Irreducible Factors

Example: Evaluate $\int \frac{(x+1)(4x+1)}{(x^2+4)^3} dx$.

Eighth Idea: Combining All of the Other Ideas

Example: Evaluate $\int \frac{x^2}{x^3(x^2 + 1)} dx$.